

Plumb — Features & Functions Catalog

Plumb — Complete Features & Functions Catalog

For: Technical Architect · **Source:** Plumb Master Plan (461-line synthesis of 9 specialist agents) · **Version:** 1.0

This catalog enumerates every function the Plumb platform must deliver, organized by product, with dependencies, data touched, and release phase. Use it to scope architecture, estimate effort, and assign workstreams.

How to read this document

Column	Meaning
ID	Stable identifier — reference in tickets, designs, and code (F-01-01)
Function	What the system does
Data	Core entities touched
Phase	P0 (built) · P1 (MVP) · P2 (Beta) · P3 (GA) · P4 (Scale)

1. Plumb Capture (360° reality capture)

ID	Function	Details	Data	Phase
F-01-01	Multi-source capture	Smartphone, Insta360 X4, Ricoh Theta Z1, Matterport Pro3, GoPro MAX, iPhone Pro pano, drones, laser scans	Capture, CaptureFrame	P1
F-01-02	Auto-mapping to plans	Image-to-plan alignment: DINOv2 embeddings + PnP with floorplan priors + visual relocalization against prior captures	Floorplan, CaptureFrame, Pose	P1
F-01-03	On-device redaction	Face/plate blur at capture time (Apple Neural Engine / NNAPI)	Capture, Asset	P1
F-01-04	ARKit/ARCore visual relocalization	Anchor captures to physical space using device AR	Pose, Capture	P2
F-01-05	3D Gaussian Splatting rooms	Photoreal novel-view from sparse captures; per-room 10–50 MB	3DGS Asset	P3
F-01-06	QuickCodes	Physical QR codes that auto-tag floor/area on capture	Capture, Floorplan	P1
F-01-07	Coverage confirmation	Real-time coverage % vs plan during walk; coverage heatmap	Capture, Pose	P1
F-01-08	Split View / Reveal Mode	Two-pane capture-vs-BIM sync; reveal slider wipe	Capture, BIMModel	P1

ID	Function	Details	Data	Phase
F-01-09	Timeline scrub	Scrub through 6 months of captures per space	Capture	P1
F-01-10	Offline capture + resumable upload	SQLite-first, chunked + resumable + content-hash dedup, BGTaskScheduler/WorkManager	Capture, SyncState	P1
F-01-11	Capture pipeline (DAG)	Decode → SuperPoint → SfM (GLOMAP) → MVS (OpenMVS) → Poisson mesh → 3DGS → SAM-2 → BIM alignment, with backpressure + DLQ	PipelineRun, Artifact	P1

2. Plumb Field (field notes & issues)

ID	Function	Details	Data	Phase
F-02-01	Rich-text field notes	Text + photo + sketch + voice memo + attachments	FieldNote, Attachment	P1
F-02-02	On-device Whisper dictation	ASR offline; auto floor/area tag via indoor positioning	FieldNote	P2
F-02-03	Sketch-on-BIM	Draw annotations directly on BIM model view	Annotation	P2
F-02-04	Issue tracking board	Kanban + list; severity, assignee, due, area, type filters; drag-to-reassign	Issue, IssueComment	P2
F-02-05	Photo with QuickCode	Auto-tag location from QR; upload with note	Attachment	P1
F-02-06	Offline-first sync	LocalID (UUIDv7) → ServerID; idempotent via clientId + contentHash; Lamport timestamps	FieldNote, SyncState	P1
F-02-07	Indoor positioning	Wi-Fi RTT + BLE + UWB + AR visual loc → automatic floor/area tagging	Pose	P2
F-02-08	Voice memos	Record, transcribe (server or device), attach to note	Attachment	P1
F-02-09	Punch list / task assignment	Assigned tasks with one-tap photo proof (subcontractor foreman persona)	Issue, Task	P2

3. Plumb Track (progress tracking)

ID	Function	Details	Data	Phase
F-03-01	BIM-vs-reality percent complete	Per-trade / floor / area with β -distributed confidence intervals ($\pm 3\%$ abs, $\pm 5\%$ rel vs hand count)	ProgressSnapshot	P2
F-03-02	EV S-curve & Gantt	Earned-value curves, schedule Gantt on canvas	ProgressSnapshot, Schedule	P2
F-03-03	Trade $\times$ floor heatmap	Visual percent-complete matrix	ProgressSnapshot	P2
F-03-04	Schedule risk forecast	Risk vs impact scatter; weather impact signals	Schedule, Risk	P2
F-03-05	Cost-code rollup	Progress $\rightarrow$ cost code mapping for estimators	CostCode	P3
F-03-06	Auto-quantities	Quantities from aligned captures + BIM for change orders	Quantity	P3

4. Plumb Air (drone coordination)

ID	Function	Details	Data	Phase
F-04-01	Drone capture ingest	DJI / Arducopter / Auterion SDK; PPK/RTK georeferencing	Capture	P2
F-04-02	Photogrammetry	OpenDroneMap: orthomosaic + DSM + point cloud	Asset, Ortho	P2
F-04-03	Volumetric/area measurement	Measure stockpiles, cut/fill from ortho + DSM	Measurement	P2
F-04-04	LAANC integration	Airspace authorization at flight-planning step	FlightPlan	P2
F-04-05	BVLOS-aware flight logs	FAA Part 107 compliance logging	FlightLog	P2
F-04-06	Drone-in-a-Box	DJI Dock / Skydio Dock scheduled autonomous flights	FlightPlan, Capture	P4

5. Plumb Model (BIM coordination, was “BIM+”)

ID	Function	Details	Data	Phase
F-05-01	Federated BIM viewing	glTF + IFC (2x3/4/4.3) + NWD via Forge Data Exchange	BIMModel	P2
F-05-02	Clash heatmap	AI-ranked clash detection (BIM vs BIM, BIM vs point cloud)	Clash	P2
F-05-03	BCF 3.0 import/export	BCF topics round-trip with other tools	BcfTopic	P2
F-05-04	Design-intent delta diff	Compare design vs as-built delta	DeltaDiff	P2
F-05-05	AR overlay on site	BIM overlaid on live camera; see-through-walls verification	BIMModel, Pose	P3
F-05-06	Generative base model	Auto-generate base BIM from floorplan when no model exists	BIMModel	P3
F-05-07	Section planes & measurement	Slice model; measure in 3D	Measurement	P2
F-05-08	2D/3D side-by-side	Sheet vs model comparison	BIMModel, Sheet	P2

6. Plumb Copilot (AI assistant)

ID	Function	Details	Data	Phase
F-06-01	Function-calling LLM	Tools: query captures, spatial query, diff captures, create issue, generate report, draft RFI, summarize progress, measure	ProjectGraph	P2
F-06-02	Citations mandatory	Refusal-on-no-citation; every claim resolves to a project artifact	Citation	P2
F-06-03	RAG over project graph	Hybrid BM25 + dense (pgvector/OpenSearch); Embedding reranker		P2
F-06-04	Streaming chat UI	SSE deltas; tool-call UI; full-screen chat + canvas	—	P2
F-06-05	Voice mode	Whisper ASR + TTS; barge-in	—	P3

ID	Function	Details	Data	Phase
F-06-06	Safety pipeline	PII redactor → prompt firewall → context quarantine → tenant- pinned inference; content classifier → PII inverse → C2PA provenance	AuditLog	P2
F-06-07	Report generation	Auto board decks, RFI packs, change-order bundles	Report	P2
F-06-08	Regional inference	EU/JP/KSA data- residency inference (SageMaker/Llama on- prem)	—	P4

7. Plumb Voice (hands-free)

ID	Function	Details	Data	Phase
F-07-01	“Hey Plumb” wake word	On-device, low-power always-listening	—	P1 (EN) / P2 (multi)
F-07-02	On-device ASR	Whisper tiny/base; offline dictation	—	P1
F-07-03	Multilingual	EN/ES/FR/DE/PT/HI/ZH at launch	—	P2
F-07-04	Speech-to-intent	Field commands: “log issue”, “start capture”, “find RFI”	Intent	P2

8. Plumb Live (multi-stakeholder walkthroughs)

ID	Function	Details	Data	Phase
F-08-01	Multi-user 360 room	Owner/GC/design/lender/insurance in same walkthrough	Presence	P4
F-08-02	Pass-the-pointer	Shared pointer + camera sync; voice/video/pin overlay	Presence	P4
F-08-03	Auto-minutes	Recorded → Transcribe → LLM summary	Minutes	P4
F-08-04	Realtime stack	Phoenix Channels + Redis Pub/Sub + WebRTC (IVS)	Presence, Stream	P4

9. Plumb Twin (digital twin)

ID	Function	Details	Data	Phase
F-09-01	Continuous delta detection	Every capture → diff vs prior state	DeltaDiff	P4
F-09-02	Handover package	Federated model + capture history + O&M + ESG export	Package	P4
F-09-03	FM integrations	Akila, Willow, AWS IoT TwinMaker, Azure Digital Twins	TwinLink	P4
F-09-04	Live state model	Queryable as-built state per element	ElementState	P4

10. Plumb ESG

ID	Function	Details	Data	Phase
F-10-01	Embodied carbon tracking	BIM material quantities → CO2e	Carbon	P4
F-10-02	Waste tracking	Construction-phase waste streams	Waste	P4
F-10-03	Credit readiness	LEED / Envision / BREEAM checklists	Credit	P4
F-10-04	Disclosure reporting	GRESB / MSCI / SEC climate disclosure packs	Report	P4

11. Plumb Claims (legal-grade capture)

ID	Function	Details	Data	Phase
F-11-01	Chain-of-custody	RFC 3161 timestamps on every artifact	Signature	P4
F-11-02	Notarization	Notarized capture records	Notary	P4
F-11-03	Dispute-ready export	Single-PDF evidence pack; expert-witness template	Export	P4

12. Plumb Edge (on-device AI)

ID	Function	Details	Data	Phase
F-12-01	Real-time scene understanding	On-device semantic segmentation (ANE/NNAPI)	Mask	P1

ID	Function	Details	Data	Phase
F-12-02	Local mesh stitching	On-device capture → mesh without server	Mesh	P2
F-12-03	Offline speech-to-intent	Intent parse without network	Intent	P2
F-12-04	PPE/safety detection	On-device PPE + fall-hazard detection; auto-flagged captures	Detection	P1

13. Plumb Admin & Trust

ID	Function	Details	Data	Phase
F-13-01	Org/portfolio dashboard	KPI tiles, activity stream, org switcher	Org	P1
F-13-02	RBAC + ABAC	Org roles × project roles; attribute-based policies; per-tenant isolation	Role, Policy	P1
F-13-03	Federated SSO + SCIM	OIDC + SAML + SCIM 2.0 provisioning	Identity	P2
F-13-04	Audit log	Append-only, hash-chained, WORM export; every state change	AuditLog	P1
F-13-05	Data residency	US/EU/UK/AU/JP/KSA pinning; region-pinned storage + inference	Region	P1
F-13-06	BYO-storage + CMK	Customer S3 + KMS keys; envelope encryption	Key, Storage	P3
F-13-07	Billing	Seat + storage + AI-minute metering; per-GB overage	Invoice	P2
F-13-08	Compliance evidence	SOC 2 / ISO 27001 / ISO 27701 / FedRAMP Moderate continuous collection	Evidence	P2+
F-13-09	MFA + adaptive risk	TOTP/FIDO2; step-up on sensitive ops	Session	P2
F-13-10	Share links	Signed, scoped, expiring tokens for read-only external views	ShareToken	P1

Cross-cutting systems (not a product, but required)

ID	System	Functions	Phase
X-01	API surface	REST (public, Idempotency-Key + RFC 7807) · GraphQL (web BFF) · gRPC (internal) · WebSocket (realtime) · webhooks (partners, HMAC, replay) · OpenAPI 3.1 + Protobuf + SDL in CI	P1
X-02	Ingestion pipeline	Chunked resumable upload → virus scan → transcode → SfM → tile pyramid → segmentation → embedding → BIM alignment → publish (DAG w/ DLQ)	P1
X-03	Storage	S3 bucket layout: raw-360, derived-tiles, bim-tessellations, drone-geo, thumbnails, exports; lifecycle → Glacier; CRR only where residency allows	P1
X-04	Search	OpenSearch; per-tenant index aliases; hybrid BM25+dense	P2
X-05	Realtime	WebSocket topics per project; presence; soft-realtime SLA; SSE fallback	P1
X-06	Notifications	In-app + email + Slack/Teams + push (APNs/FCM); Live Activities	P1
X-07	Integrations (16)	Procore, ACC/BIM 360, P6, MS Project, Salesforce, ServiceNow, Slack, Teams, Box, GDrive, DocuSign, Smartsheet, Bluebeam, Aconex, Outlook, GCal	P2 (v1: Procore/ACC/P6)
X-08	Billing	Metering → invoicing; usage dashboards	P2
X-09	Feature flags	Unleash/Flipit; progressive rollouts	P1
X-10	Mobile BFF / Web BFF	BFFs per surface; GraphQL for deep-link views	P1

Core data model (18 entities)

Every entity carries region + tenant_id (data-modeled tenancy — a core architectural invariant).

Release phasing summary

Phase	Scope	Exit criteria
P0 (done)	Monorepo, tokens, org/project/user/membership services, auth, observability, dashboard shell, CI	114 tests green, 3 commits
P1 MVP	Capture (360+smartphone+QuickCodes), Split View, Reveal, Coverage, Field (text/photo/voice), Voice EN, Edge v1 redaction, Project Hub, auth, ingest DAG	TTF capture < 5 min, 5 paid, NPS > 30
P2 Beta	Field full (dictation/sketch/auto-floor), Issue board, Track v1, Model v1, Air v1, Voice multilingual, Integrations v1, Copilot v1 (chat+citations), SOC 2 prep	50 paying projects, NPS > 45
P3 GA	All core + Voice + Edge GA, Copilot voice, 3DGS rooms, AR overlay, FedRAMP Moderate, pricing live	GA launch
P4 Scale	Live, Twin, ESG, Claims, Drone-in-a-Box, on-prem/air-gapped, regional inference, CMMC/HIPAA	5,000 projects, NPS > 60, NRR > 130%

Appendix: Performance & quality budgets the architect must design for

- Cost-per-capture ≤ **\$15 fully-loaded** (compute \$13 + storage \$0.40 + LLM \$0.85 + telemetry \$0.25) at scale
- ≤ 30 GPU-min per 10K-frame capture; ≤ \$0.50/capture amortized
- First-preview 360 ≤ 10 min; photoreal novel-view p50 ≤ 1.3 h, p95 ≤ 3 h
- p95 dashboard TTI < 1.5 s; 360 first-pixel < 800 ms; 60 fps M1 / 45 fps Intel Iris Xe
- ≥ 95% field capture success offline; 100% sync on reconnect
- SLOs: API p99 < 500 ms; availability 99.9%; capture ingest 99.5%
- Multi-tenant isolation tested across ALL data planes (search, vectors, caches, analytics, logs, exports)